

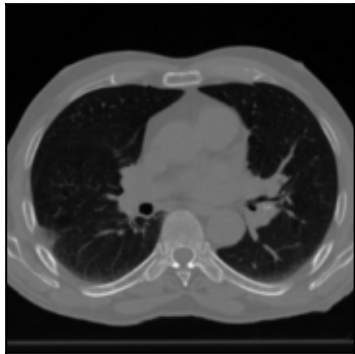
# DVF QA pipeline report

DIR-Lab-case3 [shroud\_lower]

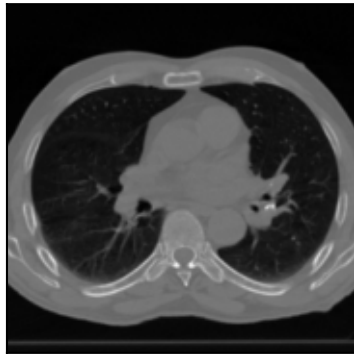
| Field            | Value                        |
|------------------|------------------------------|
| Case             | DIR-Lab-case3 [shroud_lower] |
| Generated        | 2026-05-20 08:02:33          |
| source           | Case3Pack                    |
| lateral_strategy | shroud_lower                 |
| fps              | 15.0                         |
| n_cycles         | 4                            |
| frames_per_cycle | 30                           |
| n_phases         | 10                           |
| ct_shape         | (104, 256, 256)              |
| spacing_xyz_mm   | (1.15, 1.15, 2.5)            |
| detector_shape   | (256, 256)                   |
| pixel_spacing_mm | (1.0, 1.0)                   |

4DCT phase preview

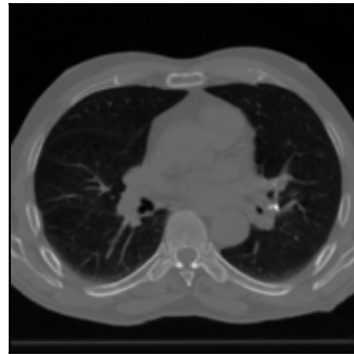
$\phi_0$



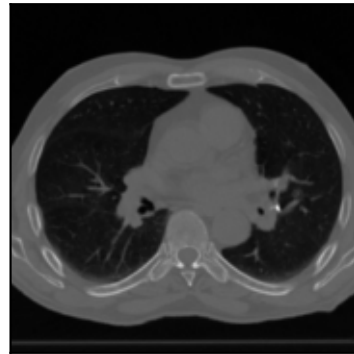
$\phi_1$



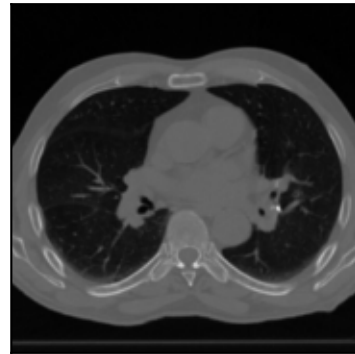
$\phi_2$



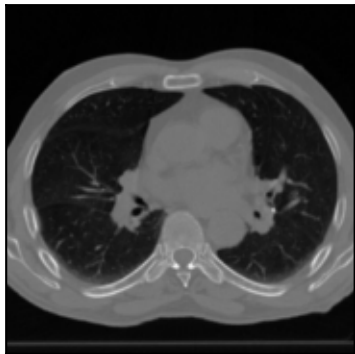
$\phi_3$



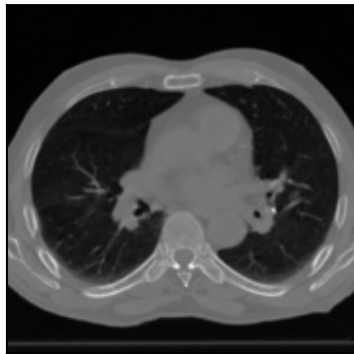
$\phi_4$



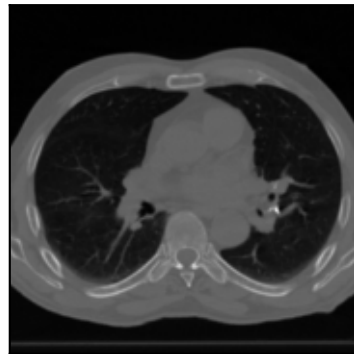
$\phi_5$



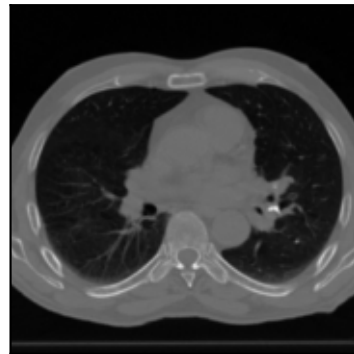
$\phi_6$



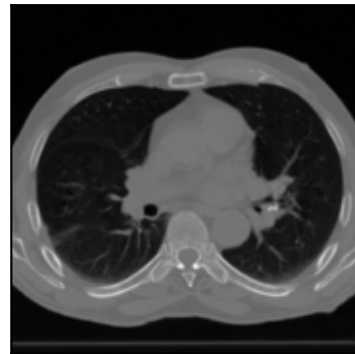
$\phi_7$



$\phi_8$



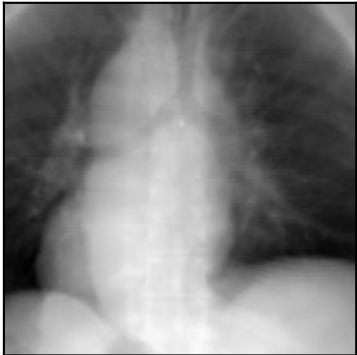
$\phi_9$



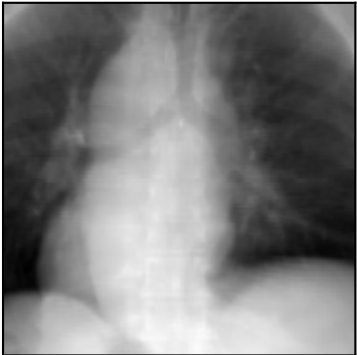
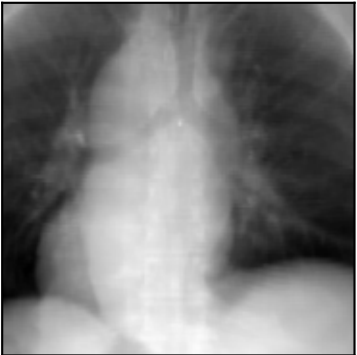
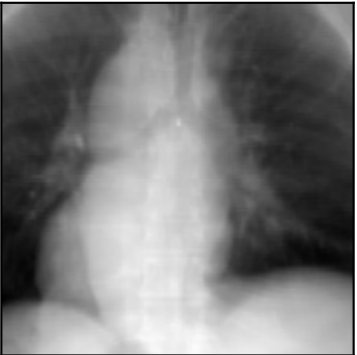
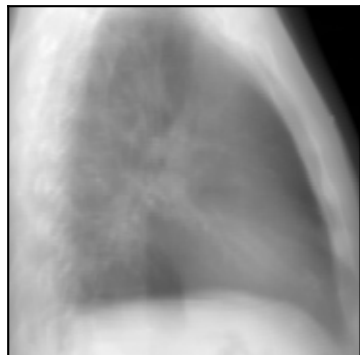
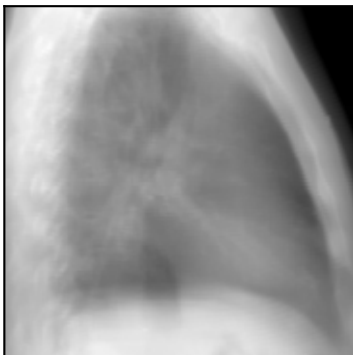
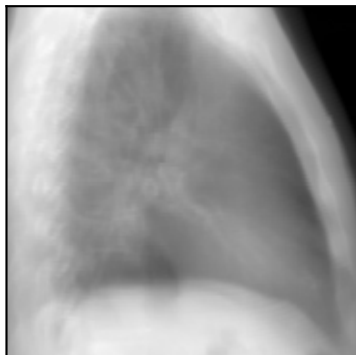
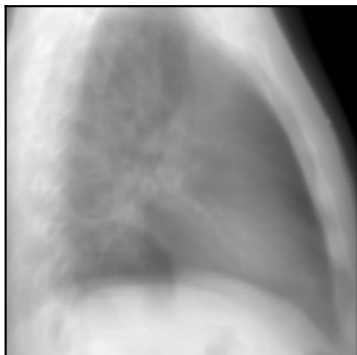
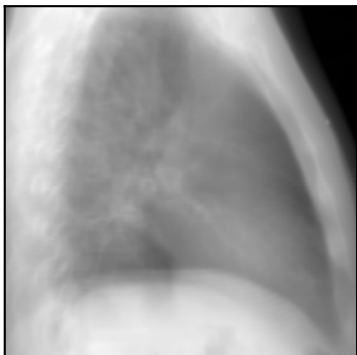
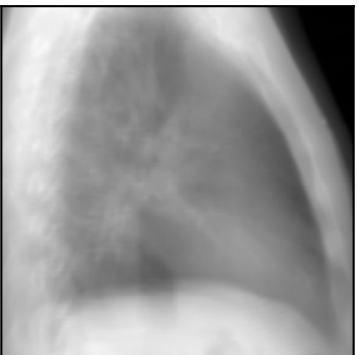
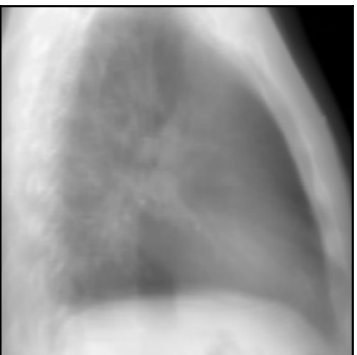
### Per-phase DRRs

AP  $\varphi_0$ AP  $\phi 1$ AP  $\phi_2$ AP  $\phi 3$ 

AP ϕ4

AP  $\phi 5$ 

AP φ6

AP  $\phi 7$ AP  $\phi 8$ AP  $\phi 9$ Lat  $\phi_0$ Lat  $\phi_1$ Lat  $\phi_2$ Lat  $\phi_3$ Lat  $\phi_4$ Lat  $\phi_5$ Lat  $\phi_6$ Lat  $\phi_7$ Lat  $\phi_8$ Lat  $\phi_9$ 

Dynamic sequence samples

AP t=0 (0.00s)



AP t=24 (1.60s)



AP t=48 (3.20s)



AP t=72 (4.80s)



AP t=96 (6.40s)



AP t=120 (8.00s)



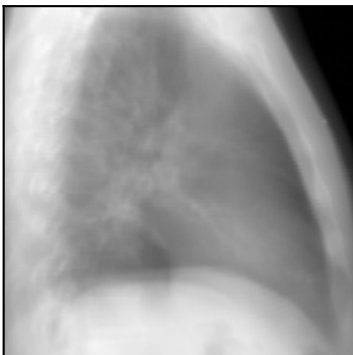
Lat t=0



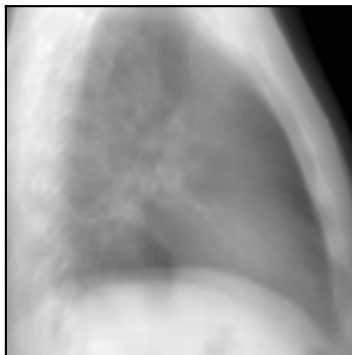
Lat t=24



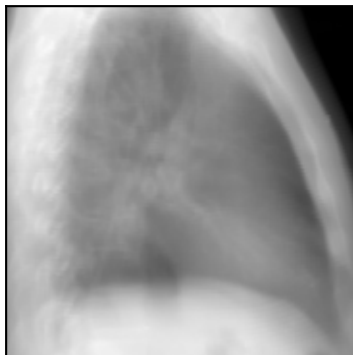
Lat t=48



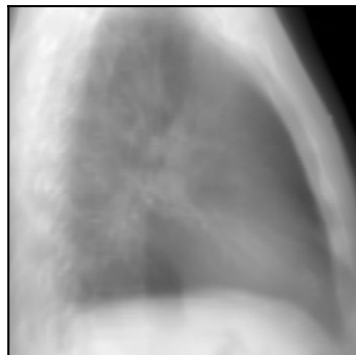
Lat t=72



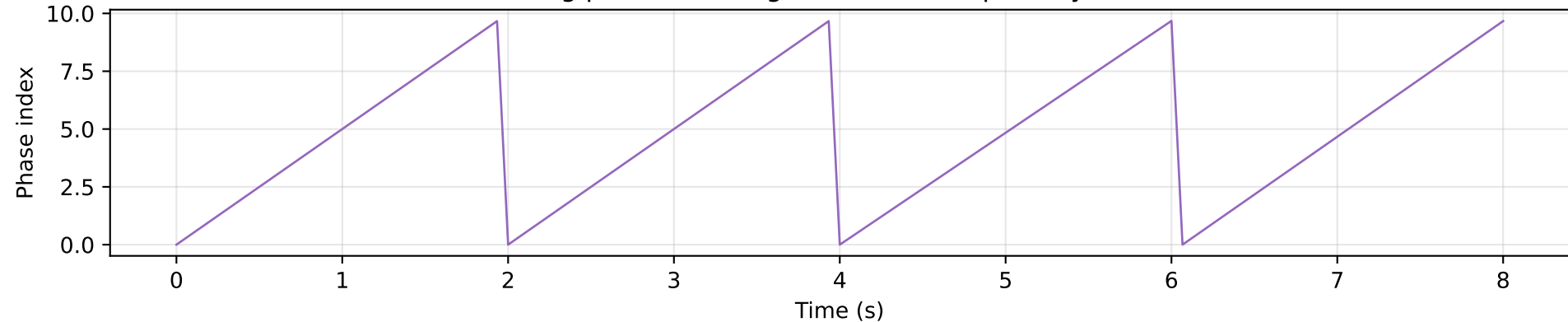
Lat t=96



Lat t=120

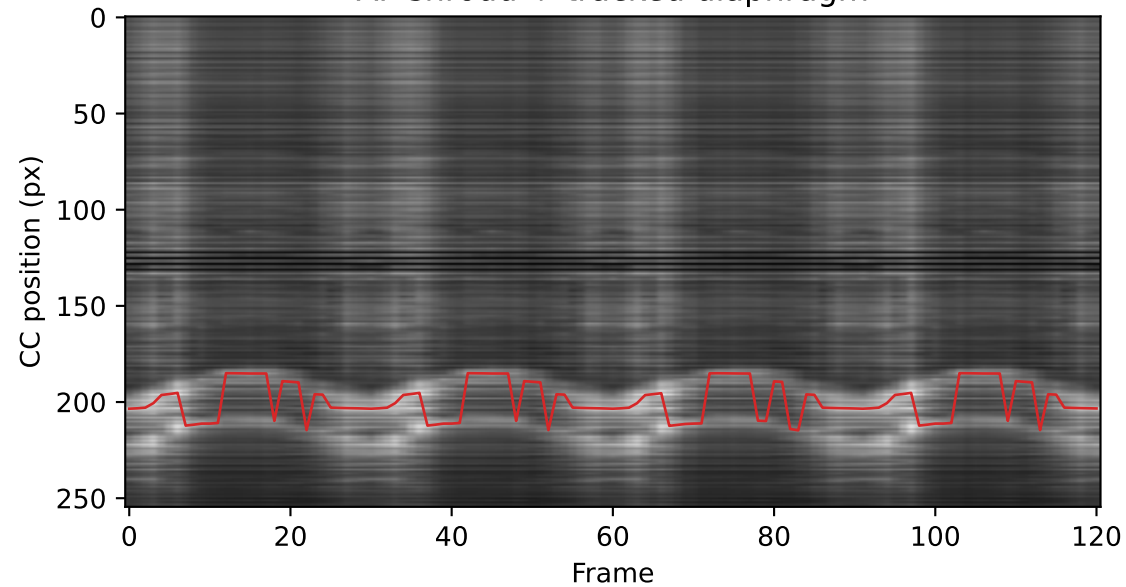


Driving phase curve  
Driving phase curve (ground-truth respiratory motion)

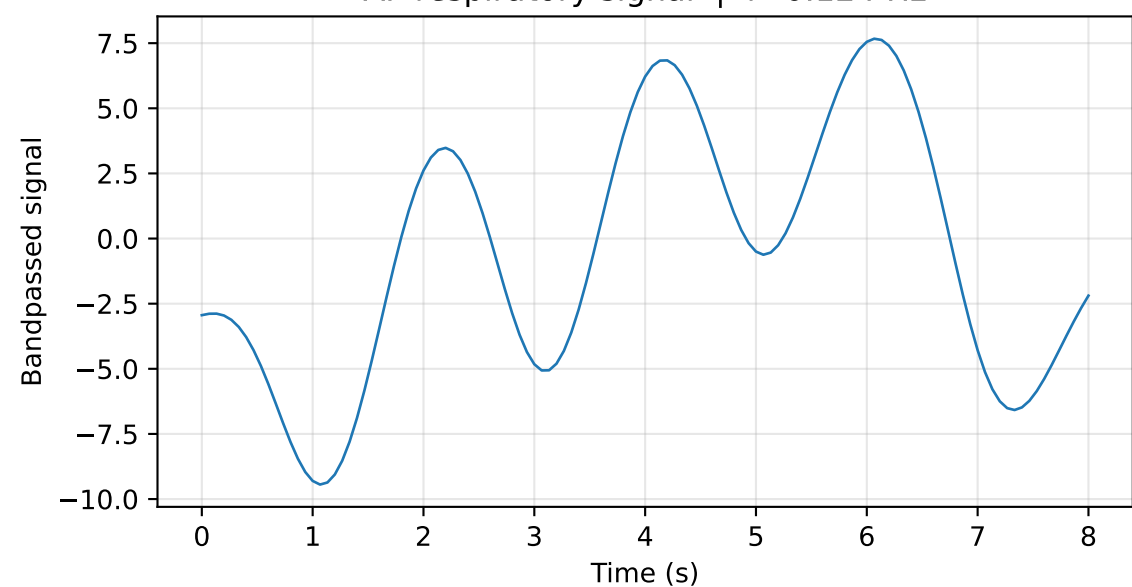


## Amsterdam Shroud — AP

AP shroud + tracked diaphragm

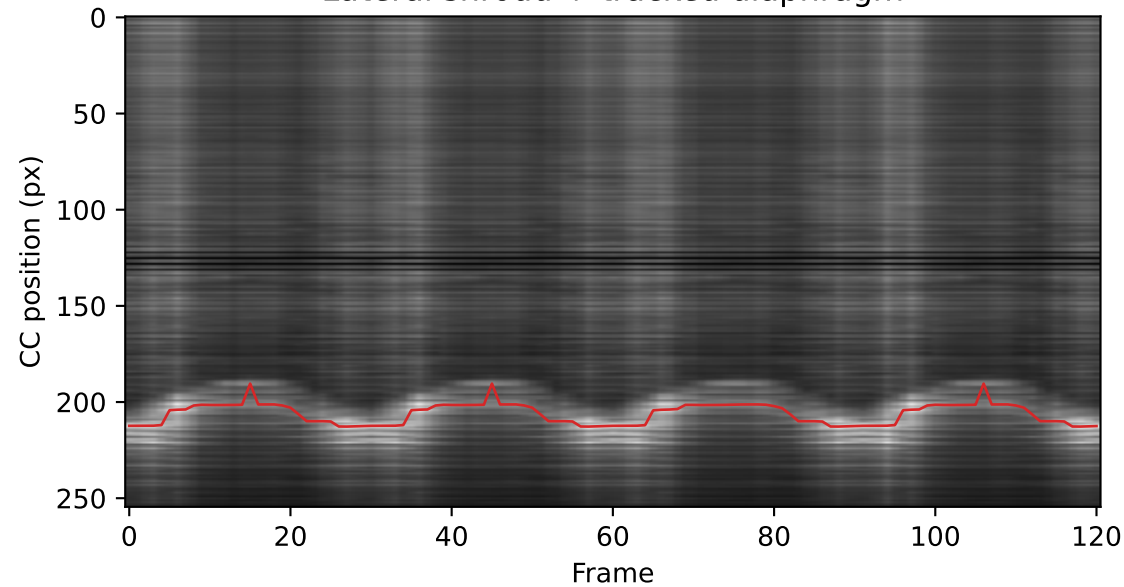


AP respiratory signal |  $f=0.124$  Hz

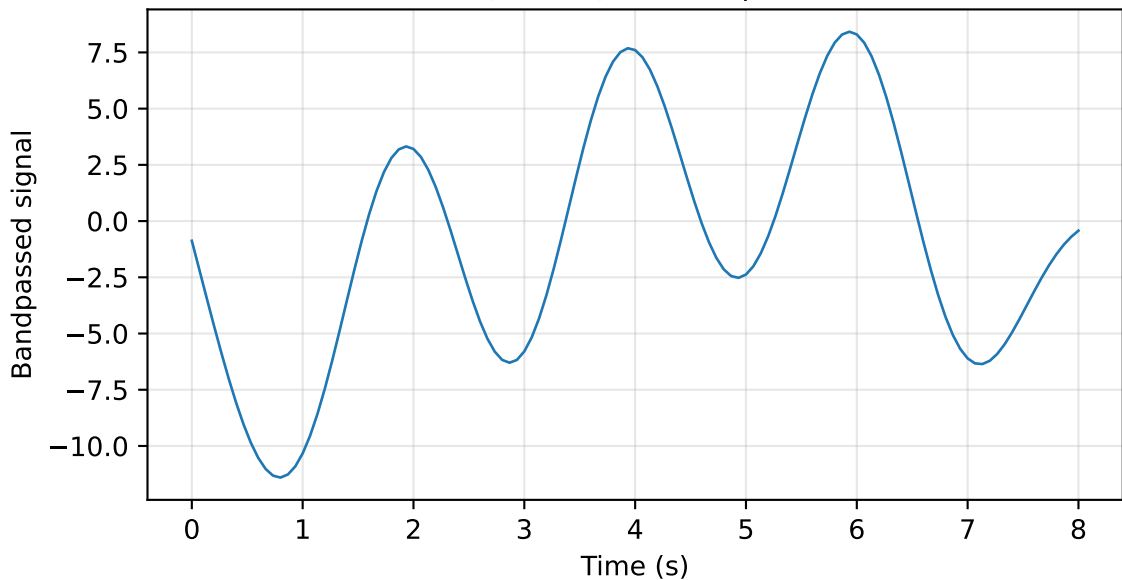


## Amsterdam Shroud — Lateral

Lateral shroud + tracked diaphragm

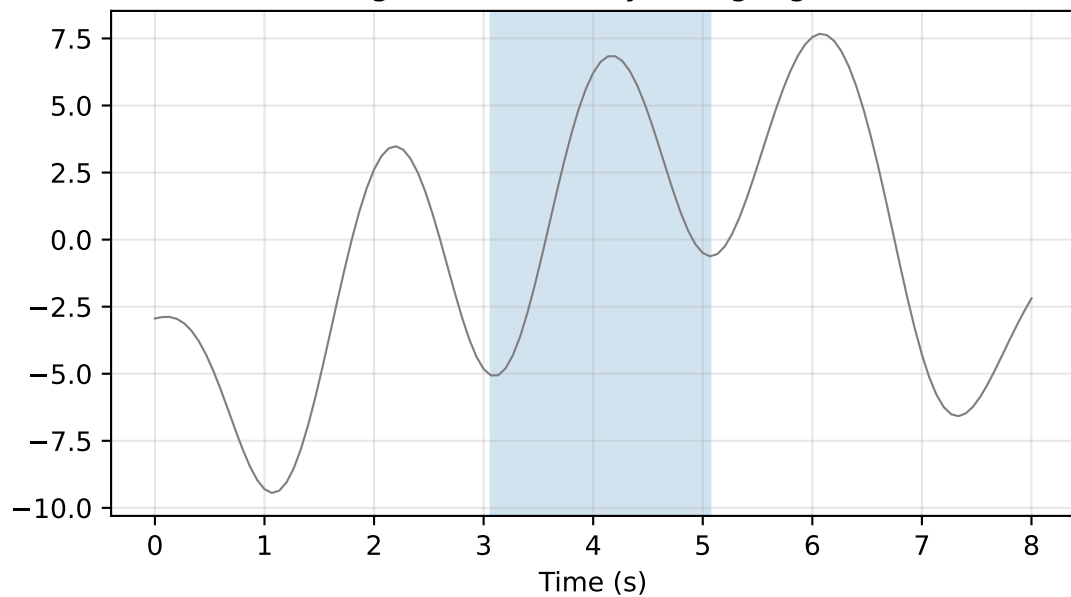


Lateral respiratory signal |  $f=0.496$  Hz

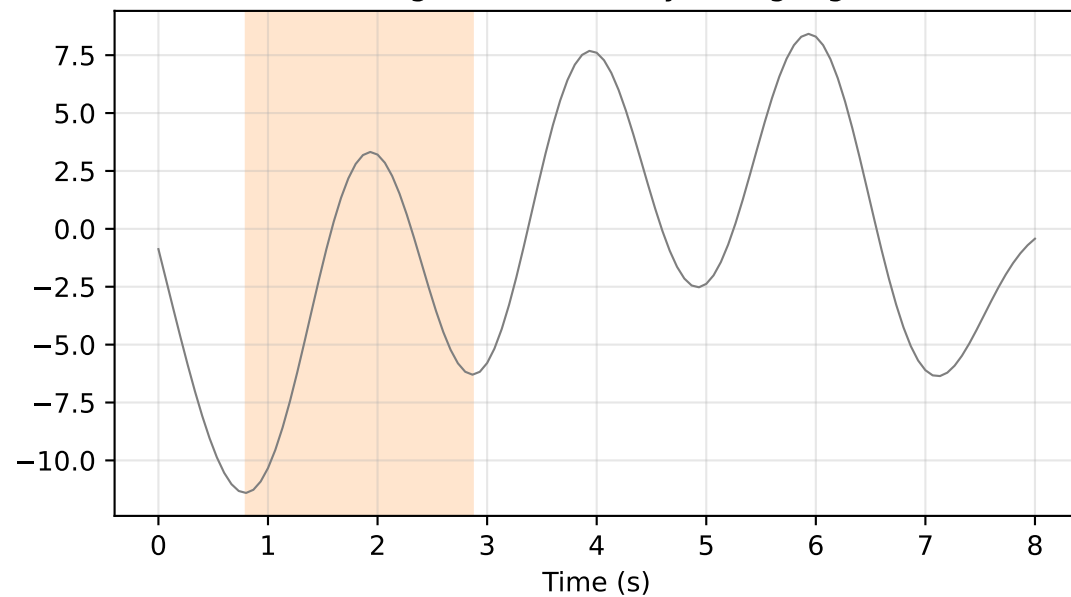


## Cycle pairing

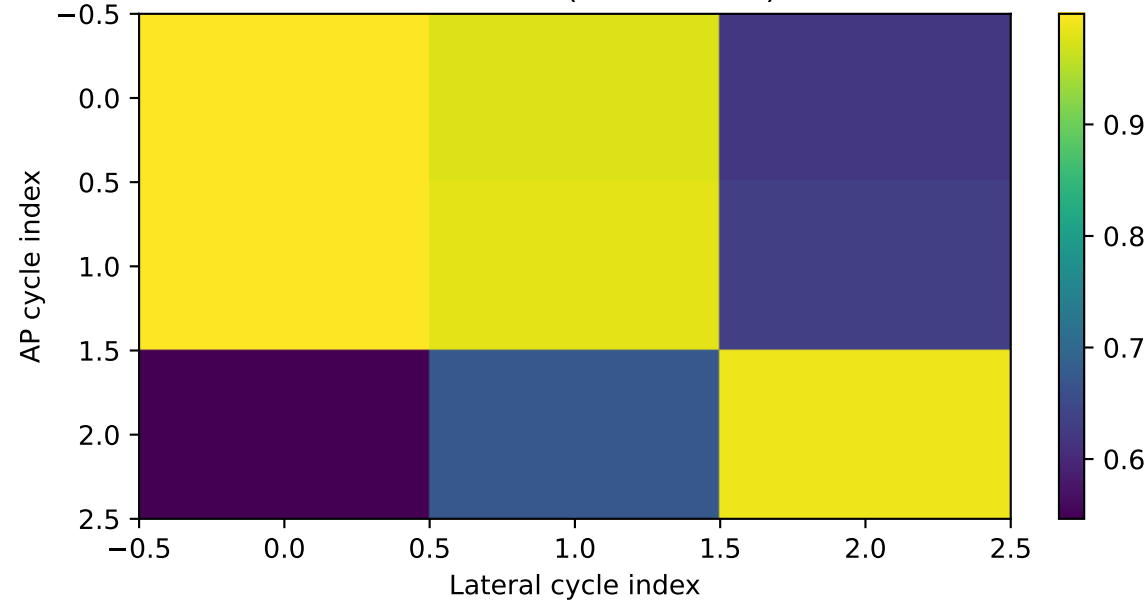
AP signal (selected cycle highlighted)



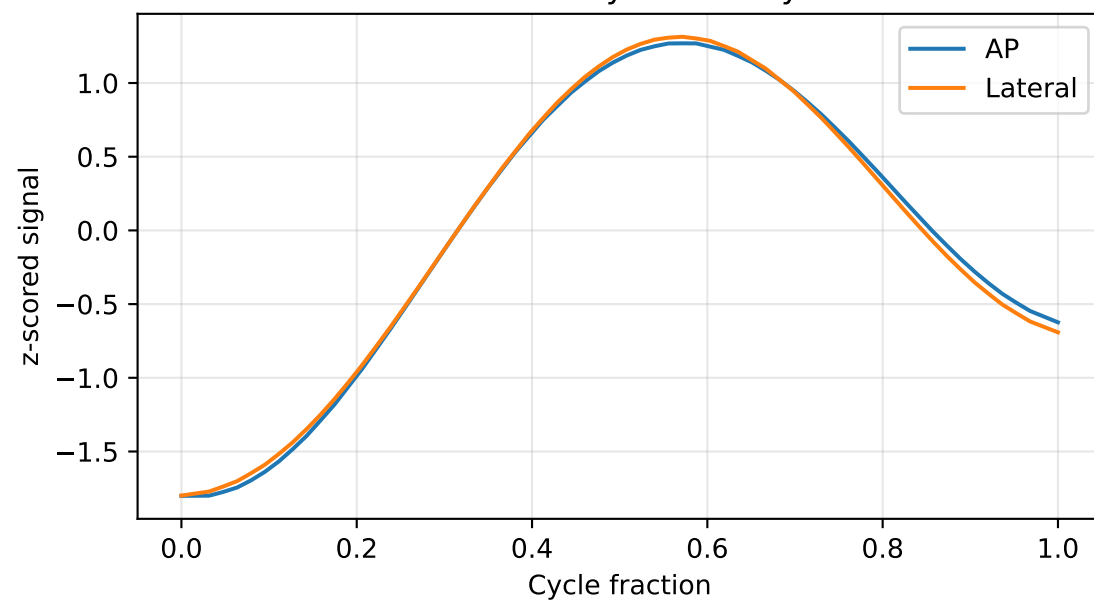
Lateral signal (selected cycle highlighted)



NCC matrix (best=0.999)



Selected cycle overlay





# Metrics

| Metric                          | Value   |
|---------------------------------|---------|
| AP dominant frequency (Hz)      | 0.1240  |
| AP period (frames)              | 121.00  |
| Lateral dominant frequency (Hz) | 0.4959  |
| Lateral period (frames)         | 30.25   |
| Best pair NCC                   | 0.9992  |
| AP cycle frames                 | 46 - 76 |
| Lateral cycle frames            | 12 - 43 |